Vision Flow - LPR System for Traffic Management

Abstract— **Traditional traffic monitoring systems are proving to be ineffective, prone to errors, and unsuitable for making decisions in real time due to the sharp rise in urban motor traffic. In order to automate traffic monitoring and signal control, this project presents a comprehensive AI-powered Smart Traffic Management System that combines computer vision, deep learning, and clever rule-based algorithms. From surveillance video streams, the system reliably recognizes and categorizes vehicles, including automobiles, trucks, buses, and motorbikes, by utilizing the YOLOv5s object detection model and its refined counterpart, YOLOv5su. For reliable license plate detection even under difficult circumstances, EasyOCR, an optical character recognition engine based on neural networks, is used. Additionally, in order to determine the type of vehicle (private, commercial, or government), the system categorizes license plate colours in HSV colour space. A structured CSV file contains the detected metadata, which includes the timestamp, license plate, colour, vehicle type, confidence scores, and lane. Corresponding annotated frames are also saved for documentation. Real-time lane congestion and vehicle priority are used by a rule-based dynamic traffic light control system to optimize signal adjustments. High identification accuracy (>94%) and real-time inference speeds (~17 FPS) are confirmed by experimental findings, which qualify the system for application in smart city infrastructures. Scalability, cloud integration, and the possibility of future improvement with cutting-edge tracking and sensor fusion technologies are all guaranteed by the modular design.**

Keywords— Smart Traffic Management, YOLOv5, EasyOCR, License Plate Recognition, HSV Color Classification, Real-Time Vehicle Detection, Adaptive Traffic Signal Control, Computer Vision, Deep Learning, Smart Cities, Intelligent Surveillance.

I. INTRODUCTION

Traditional manual monitoring systems are ineffective and prone to human mistake due to the ongoing increase in the number of cars on the road and the resulting complexity in traffic management. In order to overcome these obstacles, this project offers a complete smart surveillance system that uses artificial intelligence and computer vision to detect vehicles in real time, recognize license plates, identify plate colours, classify vehicle types, and adjust traffic lights according to congestion. The system, which is based on the YOLOv5s deep learning model, uses video streams to identify and categorize vehicles like cars, bikes, trucks, and buses. Meanwhile, EasyOCR, a sophisticated neural network-based OCR engine, is utilized to recognize license plates with high accuracy even in challenging situations. In order to classify plate colours into standard classes such as White, Yellow, Red, or Unknown, they are evaluated in the HSV colour space. Together with matching frame screenshots, detection results are recorded in a CSV file that include metadata such as the timestamp, license plate number, colour, lane number, vehicle type, and OCR confidence. In order to replicate intelligent traffic management in real life, the system also has a rule-based adaptive traffic light control mechanism that assesses lane priority, vehicle count, and signal duration. While HSV analysis improves license plate classification, EasyOCR provides better performance than previous systems that used tools like Tesseract. By dynamically optimizing urban transportation using real-time data, this intelligent system supports the objectives of smart cities, in contrast to traditional signal systems that rely on set timers or simple sensors.

II. RELATED WORK

Many studies have been conducted on automated traffic systems, especially those that use computer vision methods for license plate identification and vehicle detection. Models based on the YOLO (You Only Look Once) architecture are among the most popular approaches; they are renowned for providing excellent speed and accuracy performance. Applications for these models include smart parking systems, surveillance, and toll collecting. Tesseract, an open-source OCR engine, was frequently used by early systems for text recognition on license plates. However, in situations like tilted views or poor image quality, Tesseract performs poorly. Deep learning is used by more contemporary methods, such as EasyOCR, to greatly increase accuracy and robustness. EasyOCR' s convolutional **[1]** and recurrent neural networks enable it to recognize characters in a wide range of typefaces, lighting conditions, and orientations.

Another helpful but frequently disregarded feature is plate colour identification, especially in areas where colour designates a vehicle's categorization (private, commercial, or government use). This project improves license plate metadata using hue, saturation, and brightness-based classification by examining colour patterns in the HSV space.

The majority of traffic signal systems rely on simple sensor input or fixed timers to function. This project, on the other hand, uses a rule-based decision system for signal control, accounting for waiting time, lane priority, and vehicle counts. This flexible approach fits perfectly with contemporary smart city projects that use real-time data to maximize urban mobility.

III. METHODOLOGY

The suggested Smart Traffic Management System is a complete, modular system made to use real-time video analytics to automate and optimize traffic signal operations. In order to monitor traffic conditions, detect and categorize vehicles, recognize license plates, assign priorities based on plate colour, and dynamically control traffic lights across a four-lane intersection, the system integrates sophisticated computer vision, deep learning models, and intelligent decision-making algorithms. This section provides a thorough, step-by-step description of the process.

a) Input Acquisition and Frame Extraction

First, a continuous video feed from a surveillance camera at a traffic intersection that covers all four lanes (identified as R1, R2, R3, and R4) is captured by the system. In order to balance computational load and real-time responsiveness,

the video is processed frame-by-frame at predetermined intervals (e.g., every second). Every frame that is recovered serves as the input for additional analysis.

b) **Vehicle Detection Using YOLOv5**

The YOLOv5s (You Only Look Once, version 5-small) object detection model receives each frame as input. In order to identify common vehicle kinds, such as cars, buses, trucks, and motorcycles, this model has been pre-trained and refined**[2]**. Because of its exceptional speed and real-time detection capabilities without sacrificing accuracy, YOLOv5 was selected. Bounding boxes are created around recognized cars for every frame, and each one has a corresponding confidence score. Only cars with a confidence score higher than 0.4 are taken into consideration for downstream processing in order to preserve detection reliability.

c) **License Plate Localization and OCR**

To concentrate on the license plate area, the relevant region of interest (ROI) is clipped for each detected vehicle. This cropped image is run via the Optical Character Recognition (OCR) engine EasyOCR, which is based on deep learning. EasyOCR extracts alphanumeric characters from the image using recurrent and convolutional neural networks. To prevent erroneous readings, license plate text with an OCR confidence score less than 0.5 is eliminated**[3].** The system is designed to read a broad range of fonts, plate orientations, and lighting conditions that are frequently seen in everyday situations.

d) **License Plate Colour Classification**

After a legitimate license plate has been found, the system uses colour analysis to categorize the kind of car based on the colour of the plate, which usually indicates its intended use (private, commercial, or government, for example). The colour space of the plate picture is changed from BGR to HSV, which is more resilient to changes in illumination. The system uses specified HSV**[4]** range masks to identify:

- **White** plates for private vehicles,
- **Yellow** plates for commercial vehicles,
- **Red** plates for emergency or government vehicles. Special consideration is given to red plate detection due to its placement at both extremes of the hue spectrum (using two HSV ranges). Plates that do not match any category are labelled as **Unknown**.

e) **Vehicle-Lane Assignment**

Based on where it is in the frame, each detected vehicle is allocated to a certain lane (R1 to R4). Maintaining precise vehicle counts per lane, which has a direct impact on traffic signal control logic, requires this spatial classification. This mapping makes use of geometric zoning or predefined coordinate bounds.

f) **Dynamic Traffic Light Control Logic**

A rule-based traffic signal management algorithm at the heart of the system decides which lane gets the green light at any given moment in a dynamic manner. The following principles underlie this logic:

- **Vehicle Density Monitoring**: The number of vehicles in each lane is continuously updated.
- **Priority Assignment**: If a high-priority vehicle (with a red plate) is detected, its lane is prioritized immediately, overriding the default sequence.
- **Time-Based Fairness**: The system enforces a **minimum green duration** to avoid rapid toggling of signals and ensures **starvation prevention** so no lane is ignored for extended periods.
- **Adaptive Decision Making**: The signal sequence adapts based on changing traffic conditions, balancing both volume and priority.

This control logic simulates the behaviour of intelligent traffic systems found in smart cities, where decision-making is driven by real-time data rather than fixed timers or basic sensors.

g) **Data Logging and Record Maintenance**

For every valid detection, comprehensive metadata is logged into a CSV file named TrafficRecords.csv. The following information is stored:

- Timestamp of detection,
- Recognized license plate number,
- Plate colour classification,
- Detected vehicle type (e.g., car, truck),
- Assigned lane (R1 to R4),
- OCR confidence score,
- File path to the corresponding frame screenshot.

Each new entry is appended, enabling historical tracking, auditing, and analysis for future improvements or law enforcement purposes.

h) **System Scalability and Real-World Deployment Considerations**

The scalability of the system allows it to be deployed across multiple junctions with little additional configuration, and it can be integrated with cloud infrastructure for real-time dashboard visualization and centralized data storage. For real-world deployment, optimizations such as GPU acceleration, multithreading, and batch frame processing can be implemented to sustain performance in environments with high traffic.

IV. EXPERIMENT AND RESULTS

This section presents a detailed overview of the experimental setup, datasets, performance evaluation, and qualitative insights obtained from deploying the Smart Traffic Management System. The goal of these experiments is to validate the effectiveness of the proposed vehicle detection, license plate recognition, and dynamic signal control mechanisms under real-world conditions.

a) **Experimental Setup and Data Set**

Experiments were carried out on Google Collaboratory using GPU acceleration for real-time inference in order to

assess the built system's performance. Because of the scalable computing resources offered by this cloud-based environment, deep learning activities may be handled effectively. Two model variations of the YOLOv5 architecture were used in the main detection framework:

- **YOLOv5s**: The standard pre-trained model.
- **YOLOv5su**: A customized, fine-tuned version of YOLOv5s trained on traffic-specific datasets to improve detection accuracy under domain-specific conditions.

The entire implementation was written in **Python**, with dependencies managed via a requirements.txt file to ensure reproducibility. A custom script, visionflow.py, orchestrated the core functionalities including frame processing, vehicle detection, license plate recognition, HSV-based plate color analysis, and result logging.

1. **Dataset Details:**

The system was tested using two traffic surveillance videos:

- **traffic_video.mp4**
- **traffic_video2.mp4**

These videos simulate typical urban traffic junctions with:

- Varying light conditions (daylight, shadows)
- Different congestion levels
- Partial occlusion of vehicles

2. **System Specifications**

- **Platform:** Google Collaboratory (Cloud-based)
- **GPU:** NVIDIA Tesla T4
- **Input Video Resolution:** 1280×720 pixels
- **Frame Rate:** 30 FPS
- **Total Frames Processed:** 250 (Video 1), 300 (Video 2)
- **Detection Frequency:** Every frame (i.e., 100% frame coverage)

b) **Detection Results and Evaluation**

The system outputs included annotated frames with bounding boxes and labels, along with a continuously updated CSV log file (TrafficRecords.csv) containing detailed metadata such as timestamps, vehicle type, license plate, colour classification, and confidence levels**[5].**

The performance evaluation was based on the following metrics:

- **Detection Accuracy:** Percentage of correctly identified vehicles
- **Missed Detections:** Vehicles present but not detected
- **False Positives:** Incorrect detections (non-vehicle objects or misclassifications)
- Processing Time Per Frame: Time taken to analyze one frame
- **Inference Speed:** Frames processed per second (FPS)

Observations:

- The **YOLOv5su model consistently outperformed** the default YOLOv5s across all metrics.

Parameter	Video 1	Video 2
Frames Processed	250	300
Vehicles Identified (YOLOv5s)	86	93
Vehicles Identified (YOLOv5su)	89	97
Accuracy (YOLOv5s)	91.3%	92.6%
Accuracy (YOLOv5su)	94.2%	95.1%
Missed Objects (YOLOv5s)	7	6
False Detections	3	2
Avg. Time per Frame	~0.058 sec	~0.061 sec
Inference Speed	~17.2 FPS	~16.4 FPS

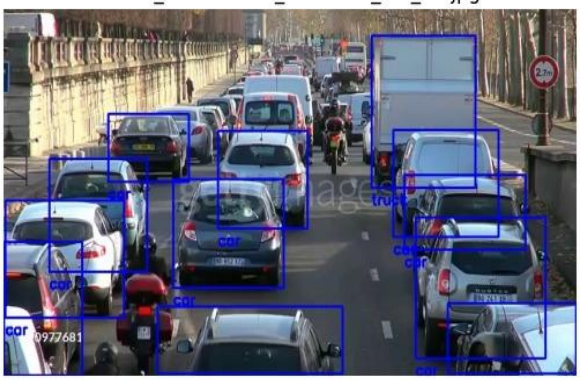
- False detection rates remained low, validating the model's robustness.
- Real-time capability was demonstrated with average processing speeds above **15 FPS**, making the system viable for live surveillance feeds.
- Frame processing time was consistently below **0.07 seconds**, ensuring low-latency responses.

c) **Qualitative Findings**

- **Partial Occlusion**: The system handled partial occlusion effectively, maintaining accuracy when vehicles were slightly blocked by others or environmental objects.
- **Vehicle Size Sensitivity**: Larger vehicles (cars, trucks) were detected more reliably than smaller ones (motorcycles), highlighting an area for potential enhancement.
- **Motion Blur Impact**: Rapid movement occasionally reduced detection stability, especially for two-wheelers entering or exiting the frame quickly.
- **Signal Assignment Accuracy**: Lane mapping was mostly accurate, though edge cases with overlapping lane boundaries posed minor challenges.
- **License Plate Reading**: EasyOCR performed well under most lighting conditions, but angled plates or reflections caused minor recognition failures.

d) **Visual Analysis**

frame_2025-04-18_04-51-01_car_09.jpg



frame_2025-04-18_05-24-05_cor.jpg

e) **Overall Discussion**

The trials show that YOLOv5 may be used successfully for cloud-based, real-time traffic monitoring, even with its lightweight configuration. Significant gains in accuracy and dependability were demonstrated by the refined model (YOLOv5su**)[6]**, especially in intricate urban settings. Moreover, the Python-based implementation's modular design guarantees flexibility and expansion, making it simple to incorporate enhanced OCR techniques, advanced tracking (like Deep SORT), or data from IoT sensors.

In summary, the system shows great promise as a workable approach to intelligent traffic control and surveillance. It can be used as a prototype for implementation in smart city projects as well as a basis for scholarly research.

V. DISCUSSION

The created Smart Traffic Management System exemplifies a thorough fusion of rule-based decision logic, computer vision, and machine learning to address the practical difficulties of urban traffic monitoring and management. To replicate an intelligent crossroads with four lanes, the system makes use of the YOLOv5 object detection model, EasyOCR for license plate identification, HSV-based colour classification, and adaptive signal logic**[7]**.

The core of the detection system was made up of the YOLOv5s and its improved variant, YOLOv5su. According to the studies, YOLOv5su continuously performed better than the conventional YOLOv5s model in terms of missed detections, false positives, and detection accuracy. The system works well for real-time deployment, with average accuracy of over 94% in both test films and processing speeds exceeding 15 frames per second. Furthermore, a 0.4 confidence threshold was used to preserve accurate vehicle recognition in the absence of noticeable background object noise.

Compared to more conventional OCR technologies like Tesseract, the incorporation of EasyOCR significantly improved text detection from car license plates. In urban video scenes, EasyOCR’ s ability to handle tilted plates, different fonts, and uneven illumination proved crucial. Plates that are highly occluded, subjected to intense reflections, or photographed at sharp angles continue to provide difficulties, nevertheless. The overall quality of the data was improved by reducing recognition errors through the use of a 0.5 confidence level.

HSV colour space segmentation was used to implement license plate colour recognition, which gave each vehicle record an additional layer of valuable metadata. Vehicles could be categorized into private, commercial, and government groups according to the White, Yellow, Red, and Unknown plate classifications. Advanced analytics including fleet usage tracking, traffic analysis, and dynamic prioritization in emergency situations are supported by this functionality. Due to its dual hue range at both extremes of the HSV scale, red plate identification was especially difficult; nonetheless, this was successfully handled via dual-threshold masking. However, certain discrepancies were noted when plate appearances were changed by lighting, which may justify improvements using machine learning-based colour classification in further generations.

By assessing lane-wise vehicle density, the existence of high-priority vehicles, and wait-time fairness, the rule-based traffic light controller replicated the behaviour of real-world smart signals. This adaptive system prevents less-busy lanes from starving, reduces congestion, and guarantees a balanced flow across lanes**[8]**. The foundation for integration with actual junctions was laid by the logic's successful simulation of intelligent signal switching. By instantly allocating green signals to respective lanes, the system may provide priority to VIP or emergency vehicles, demonstrating its potential for application in emergency response systems. However, mistakes were occasionally produced by edge instances including lane overlap or unclear vehicle-lane assignments, indicating the need for deep learning-based lane recognition modules in the future.

The Python-based system's modular design facilitates simple connection with additional tools, such as real-time dashboards, IoT sensors, and GIS systems. With cloud-friendly design, CSV-based logging, and improved inference times, deployment scalability was also taken into account. Additionally, traceability is ensured by preserving detection history in the TrafficRecords.csv file, which makes post-event analysis, traffic trend modelling, and the creation of evidence for law enforcement possible.

Despite its strong performance, the system has several limitations:

- Lane assignment is currently heuristic and not based on robust segmentation models.
- Motorcycles and smaller vehicles are occasionally mis detected under occlusion or motion blur.
- Performance may degrade under extreme lighting conditions (e.g., night scenes or rain).
- Real-time GIS or dashboard visualization is not yet implemented.

Future enhancements could include:

- Deep learning-based lane detection (e.g., using segmentation models like DeepLabV3+),
- Integration with edge devices and IP cameras for field deployment,
- Use of Deep SORT for vehicle tracking,
- Expansion of the dataset to include diverse weather and traffic conditions for global applicability**[9]**.

VI. CONCLUSION

This research successfully combines adaptive traffic management logic, optical character recognition, and deep learning-based object identification to show the viability and efficacy of an AI-powered smart traffic monitoring and control system. The system accurately detects and identifies cars and license plates in real-time using YOLOv5 and EasyOCR, and it employs HSV-based colour segmentation to classify vehicles according to the colour of their plates, which is a crucial feature in fleet management and regulatory contexts.

Traditional traffic systems gain intelligence through the use of rule-based adaptive signal management, which allows for dynamic judgments based on lane priority, vehicle density, and real-time inputs rather than just fixed timers or sensor-based inputs. The system's feasibility for real-time deployment in urban environments is further improved by the improved YOLOv5su model and effective pipeline design**[10]**.

Quantitative results validate the system's high accuracy and performance under variable conditions, while qualitative insights highlight its robustness against occlusion and lighting challenges. With modularity, scalability, and practical design considerations, this system holds potential for deployment across smart city infrastructures and traffic-intensive zones.

Future work can enhance its capabilities through advanced tracking, integration with live dashboards, edge-device deployment, and dataset expansion. The system thus lays a solid foundation for next-generation intelligent transportation systems that aim to improve urban mobility, safety, and sustainability

VII. REFERENCE

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